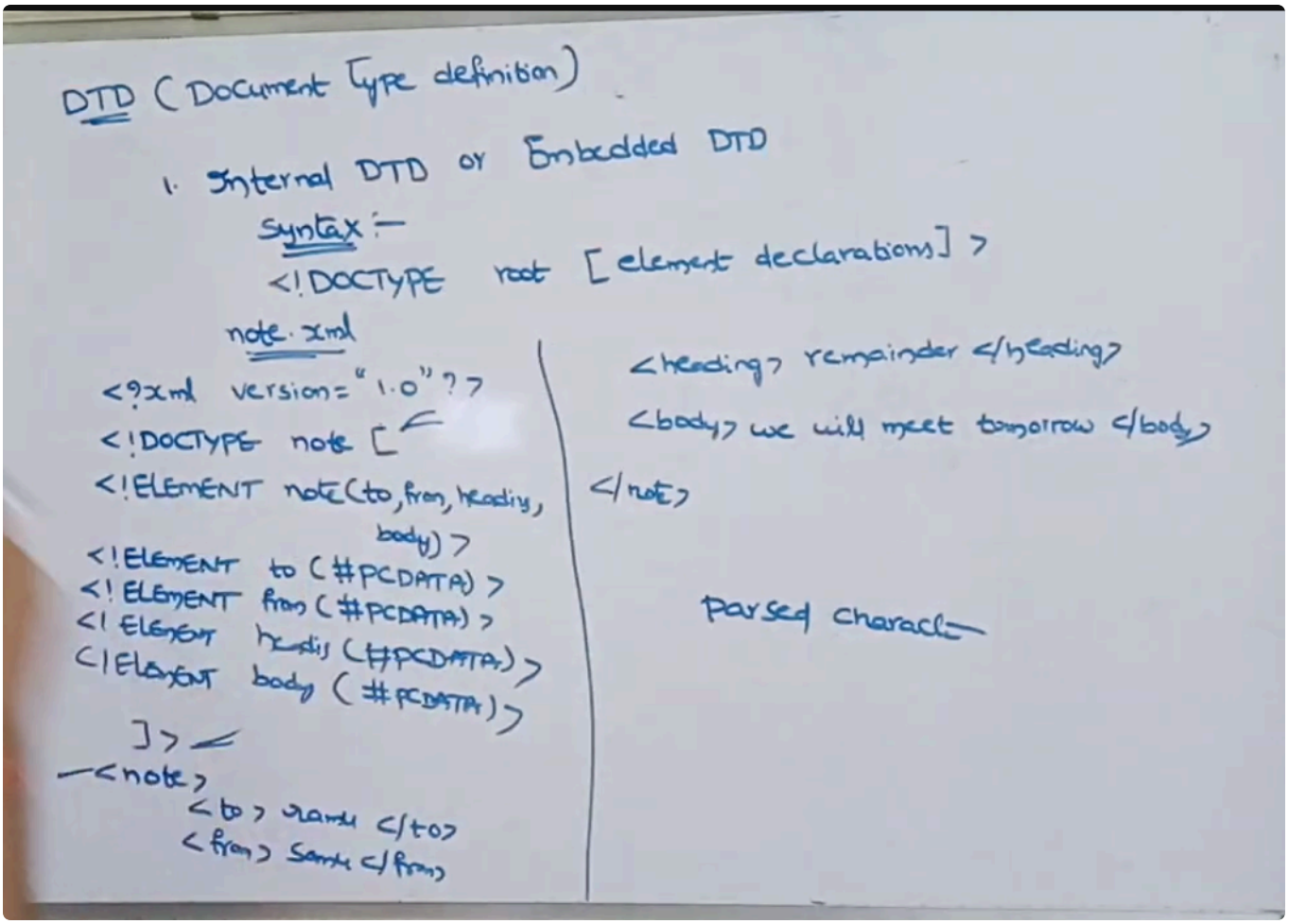
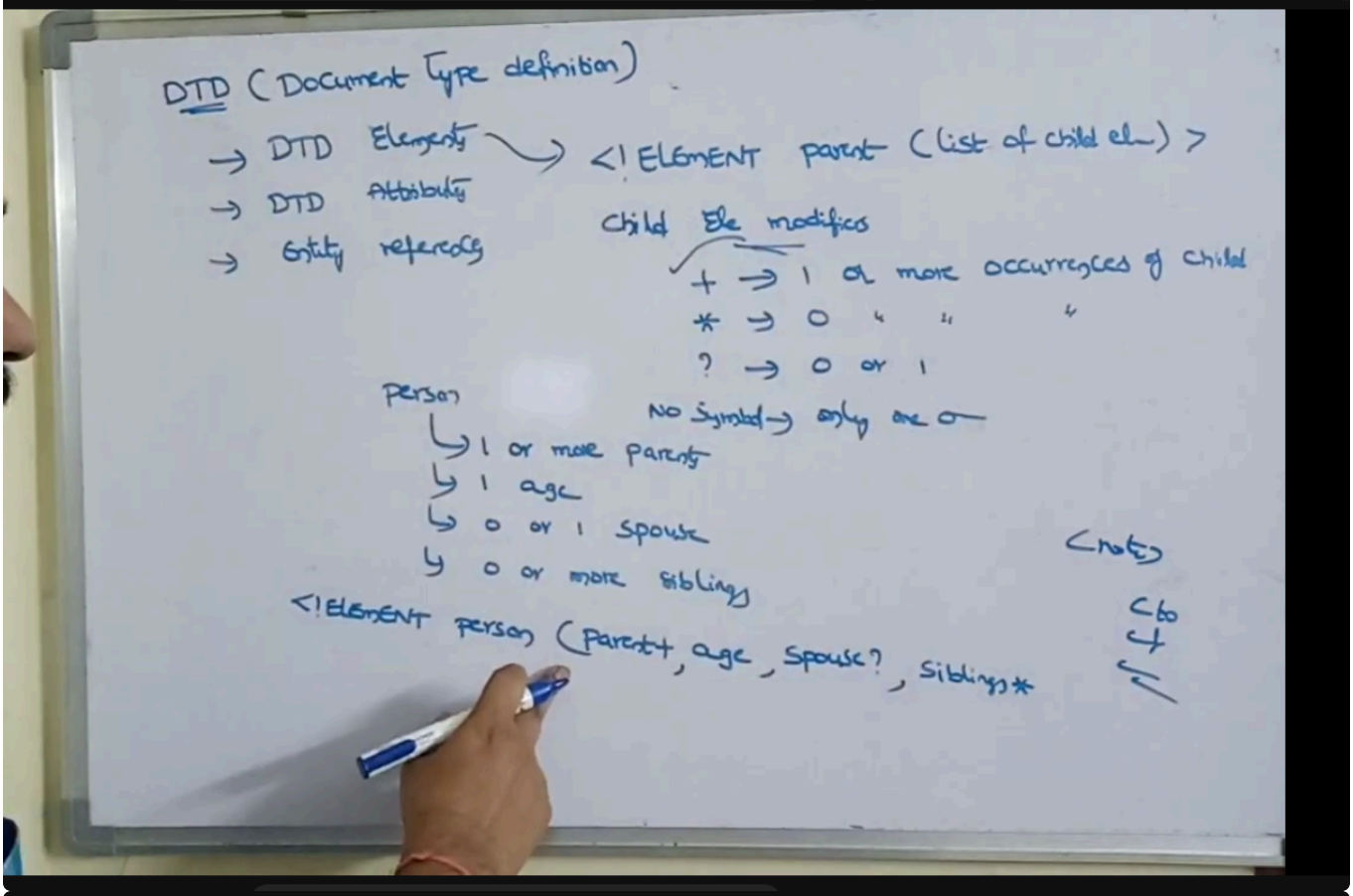
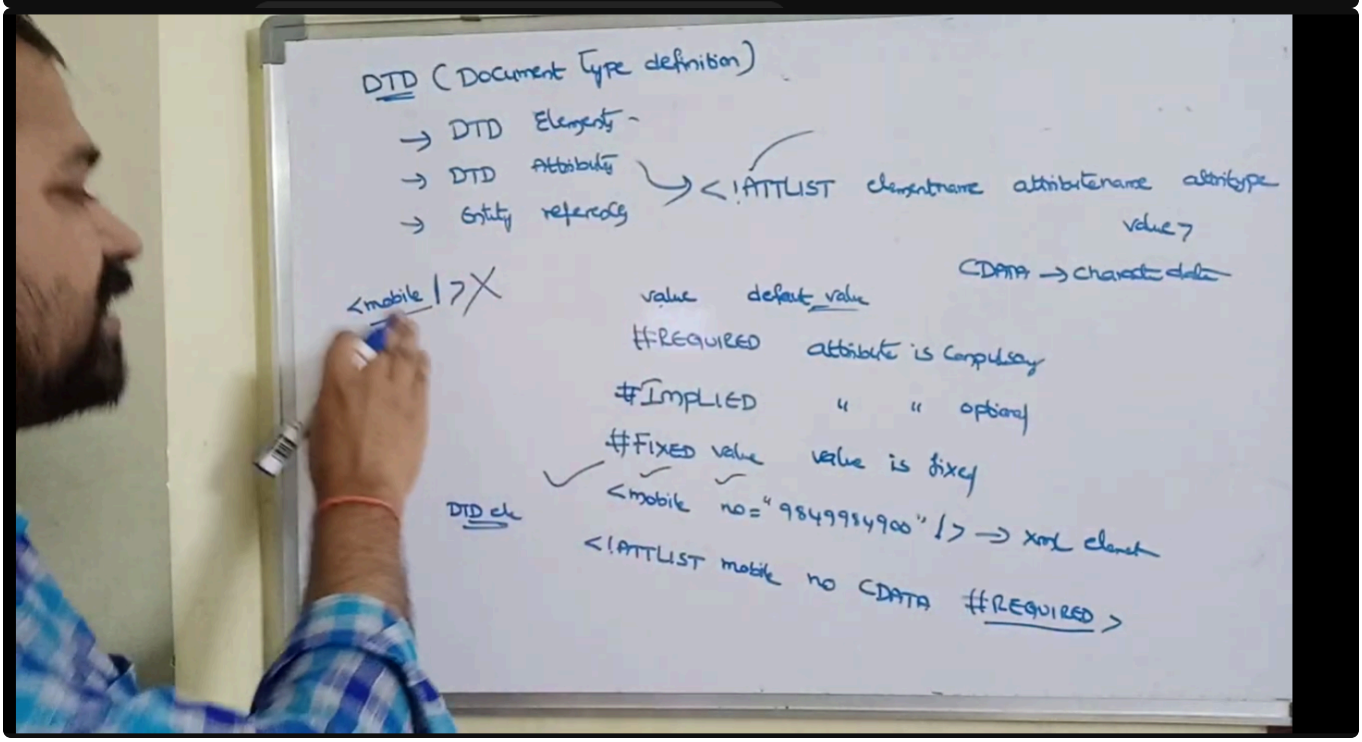
XML DTD

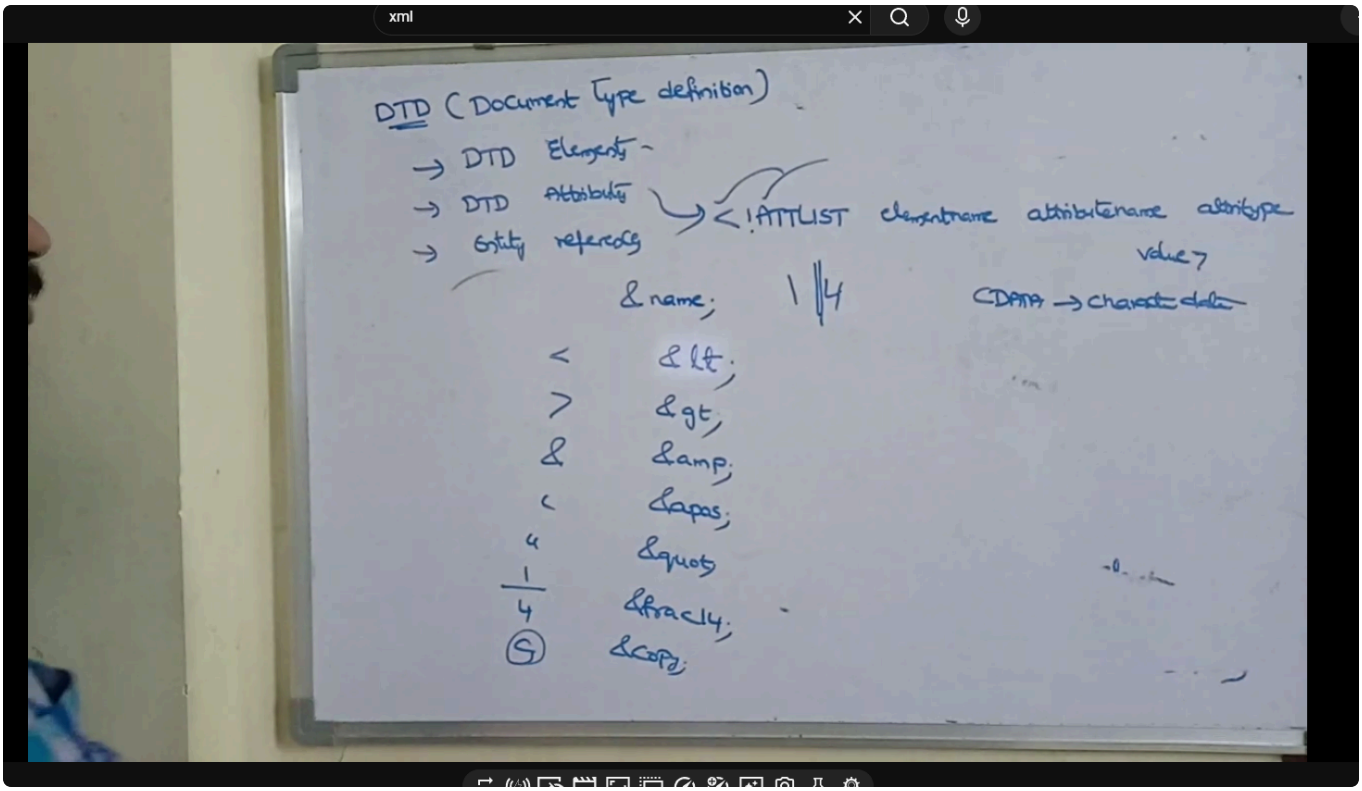
Internal and External DTD
internal DTD xml is defined in the same file
External DTD xml is defined in another dtd file and xml is one file eg: sanjay.xml and sanjay.dtd



xml dtd


DTD (Document Type definition)
→ DTD Elements → <!ELEMENT parent (list of child ele...) >
→ DTD Attributes
→ Entity reference
Child Ele modifiers
+ → 1 or more occurrences of child
* → 0 " " "
? → 0 or 1
No symbol → only one
Person
1 or more parents
1 age
0 or 1 spouse
0 or more siblings
<!ELEMENT person (parent+, age, spouse?, siblings*
<note>
<to></to>


DTD (Document Type definition)
→ DTD Elements
→ DTD Attributes → <!ATTLIST elementname attributename attributetype value>
→ Entity reference
CDATA → character data
<mobile/> X
value default value
#REQUIRED attribute is compulsory
#IMPLIED " " optional
#FIXED value value is fixed
DTD ex
<mobile no="9849984900" /> → xml element
<!ATTLIST mobile no CDATA #REQUIRED>



XML + DTD — Overall Revision for Exam

1. Basic XML structure

XML stores data using user-defined tags.

XML (`student.xml`)

```
<?xml version="1.0"?>
<!DOCTYPE student SYSTEM "student.dtd">
<student id="101">
   <name>Sanjay</name>
   <age>19</age>
</student>
```

DTD (`student.dtd`)

```
<!ELEMENT student (name, age)>
<!ATTLIST student id CDATA #REQUIRED>
<!ELEMENT name (#PCDATA)>
<!ELEMENT age (#PCDATA)>
```

- `ELEMENT` → Defines elements.
- `ATTLIST` → Defines attributes.
- `#PCDATA` → Text content inside an element.
- `CDATA` → Character data used as an attribute value.
- `DOCTYPE` → Connects XML to its DTD.

2. Element occurrence indicators

These are written in `ELEMENT` declarations.

Symbol	Meaning	Example
No symbol	Exactly once	`name`
`?`	Zero or one	`email?`
`*`	Zero or more	`subject*`
`+`	One or more	`phone+`
`,`	Sequence, in order	`(name, age)`
`|`	Choice between alternatives	`(Mr | Mrs | Miss)`

Example:

```
<!ELEMENT student (name, email?, subject*, phone+)>
```

This means:

- `name` must appear once.
- `email` is optional.
- `subject` can appear any number of times.
- `phone` must appear at least once.

3. Attribute declarations (`ATTLIST`)

Syntax:

```
<!ATTLIST element_name attribute_name type default_value>
```

Example:

```
<!ATTLIST student
   id CDATA #REQUIRED
   nickname CDATA #IMPLIED
   country CDATA "India"
   status CDATA #FIXED "Active">
```

Declaration	Meaning	XML usage
`#REQUIRED`	Attribute is compulsory	Must write it
`#IMPLIED`	Attribute is optional	Write it or omit it
`"India"`	Default value	Write it or omit it
`#FIXED "Active"`	Fixed value	Omit it or write `Active`

XML example

```
<student id="101" nickname="Sam"/>
```

Here, `id` is provided. `nickname` is optional. If omitted, `country` defaults to `India`, and `status` defaults to `Active`.

Important: Do not assign a different value to a `#FIXED` attribute.

4. Attribute types

You can restrict the kind of value an attribute accepts.

A. CDATA

```
<!ATTLIST student name CDATA #REQUIRED>
```

Accepts character data as the attribute value.

B. Enumerated values

```
<!ATTLIST student gender (Male | Female | Other) #REQUIRED>
```

XML:

```
<student gender="Male"/>
```

The value must be one of the listed choices.

5. Other element declarations

DTD	Meaning
<!ELEMENT name (#PCDATA)>	Contains text
<!ELEMENT br EMPTY>	Cannot contain text or child elements
<!ELEMENT data ANY>	Allows any content permitted by the DTD
<!ELEMENT student (name, age)>	Child elements in sequence
`<!ELEMENT payment (cash |	card)>`

6. Can child elements have attributes?

Yes. Every element can have its own `ATTLIST`.

```
<!ELEMENT student (name)>
<!ATTLIST student id CDATA #REQUIRED>

<!ELEMENT name (#PCDATA)>
<!ATTLIST name language CDATA "English">
```

XML:

```
<student id="101">
   <name language="Tamil">Sanjay</name>
```

```
</student>
```

Here, `id` belongs to `student`, while `language` belongs to `name`.

7. Common exam mistakes

- `?`, `*`, and `+` apply to element content models, not attribute declarations.
- `#REQUIRED`, `#IMPLIED`, and `#FIXED` apply to attributes.
- `,` specifies a sequence; `|` specifies alternatives.
- Child elements must follow the order defined in the DTD when a sequence is declared.
- `#FIXED` does not mean the attribute must always be explicitly written; its value is supplied if omitted.
- A DTD defines the permitted XML structure; the XML document must follow those rules to be valid.

Last-minute memory table

Keyword	Remember
ELEMENT	Defines tags
ATTLIST	Defines attributes
#PCDATA	Text inside tags
CDATA	Attribute data
#REQUIRED	Must provide
#IMPLIED	Optional
"value"	Default
#FIXED	Fixed value
?	0 or 1
*	0 or more
+	1 or more
,	Sequence
`\	`
EMPTY	No content
ANY	Any permitted content

Exam strategy: If you are given a DTD question, first identify the elements, then their attributes, then the occurrence indicators. Finally, write an XML document that satisfies all the rules.